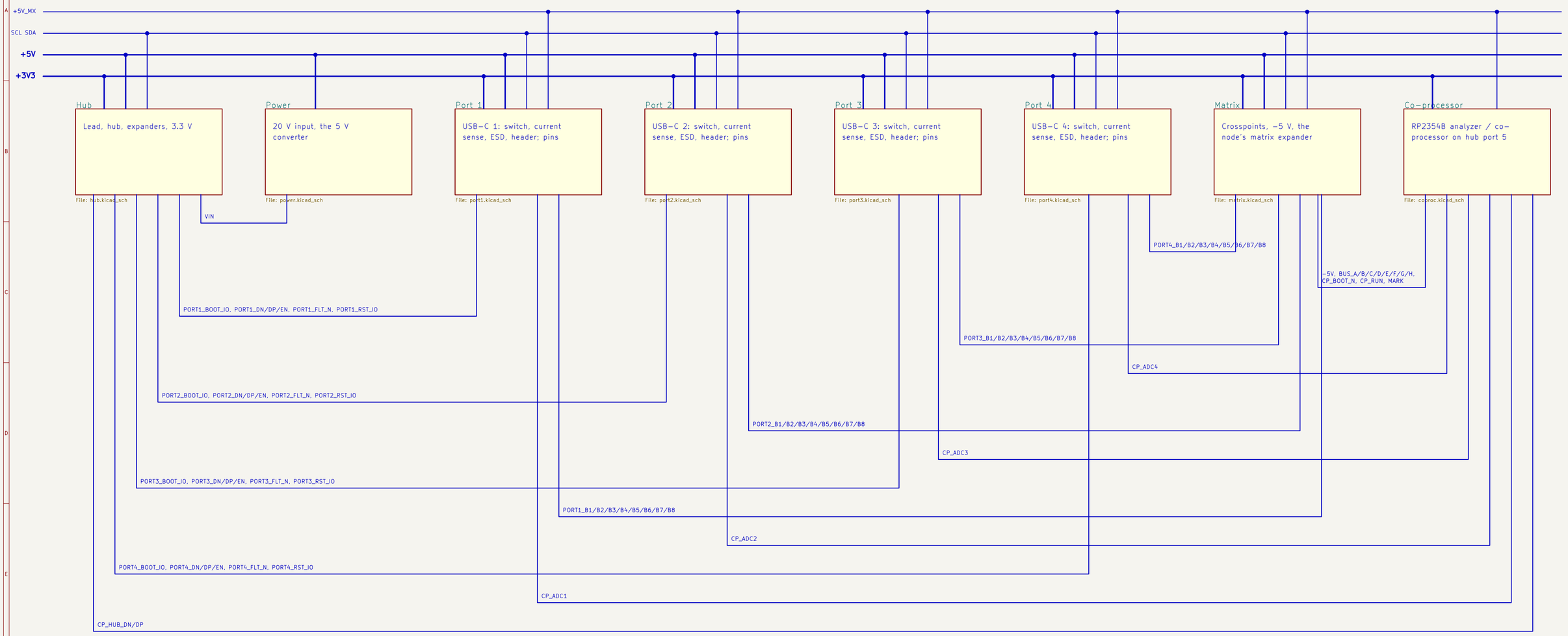


DATA ×4 v2.0 — USB2517I hub (7 ports), four USB-C (sub-slots A1.1-A1.4) each with a TPS2553-1, an INA226 and ESD, two TCA6408A for EN/FAULT/RESET/BOOT, an LMR51635 5 V from the lead's 20 V, an AP2114H 3.3 V; the PH 8 lead from the node's J7.
Pins: 32 on two CH446Q (±5 V), LSF0108 per port, an RP2354B analyzer on hub port 5, a TCA9534A for the matrix and the co-processor's RUN/BOOTSEL. Standing behind the port wall, one face assembled; v1 stays the other bays' board.
Lead pinout here: 1 NC · 2 SCL · 3 SDA · 4 GND · 5 D− · 6 D+ · 7 GND · 8 VIN — the DATA ×6's lead.
Rails: VIN 20 V (13.5-28 V; on above 13.5 V), ≤ 0.91 A at 20 V (1.21 A at 15 V) · +5V 5.24 V, ≤ 3.04 A (the ports at their 0.644 A limit + 0.46 A; the LMR51635's 3.5 A) · PORTn_VBUS 5.1-5.24 V, 0.52-0.64 A limit · +3V3 3.3 V (AP2114H), ≤ 0.5 A · +5V_MX, −5V (sheet Matrix) · +3V3_CP, +1V1 (sheet Co-processor) · HUB_1V8 / HUB_1V8PLL 1.8 V (the hub's own regulators).
I²C: TCA6408A 0x20 / 0x21, INA226 0x41 / 0x43 / 0x44 / 0x45, TCA9534A 0x38; the node's INA226 0x40.
Assembled as DATA-4x-logic (as drawn) or DATA-4x (BOM/CPL '−base': the matrix, the co-processor and the ports' pin chains not fitted).

Generated from electronics/data-port-x4-v2/design.py — edit the Python, then regenerate
A converted bay's drawer board: four USB-C behind a hub, 32 switchable pins on 8 buses, a logic an
hw-testing-rack

Sheet: /
File: data-port-x4-v2.kicad_sch

Title: DATA ×4 v2 — Overview

Size: A2	Date:	**Rev: 2.0**
KiCad E.D.A. 10.0.6		Id: 1/9

Lead from the bay node's J7 (JST PH 8, vertical, on the rear face): SDA and SCL through a H5VU25UC; pin 8 VIN (sheet Power) is the lead's 5 V.

PWR_FLAG
GND

J1
LEAD
1
2 LEAD_SCL
3 LEAD_SDA
4 GND
5 UP_DN
6 UP_DP
7 GND
8 VIN

U4
H5VU25UC
NC 10
LEAD_SDA D1+ 1
LEAD_SCL D1- 2
NC 9
GND 3
NC 7
D2+ 4
D2- 5
NC 6
GND

3.3 V for the hub, the expanders and the INA226s (AP2114H, SOT-223 on copper)

+5V
U3
AP2114H-3.3TRG1
VI 3
GND 1
VO 2
+3V3
C2
22u
GND
C3
22u
GND

VID25UH in CFG_SEL 110 (internal default configuration, straps disabled), 24 MHz crystal

+3V3
C11 100n GND
C13 100n GND
C12 100n GND
C15 100n GND
C16 100n GND
C14 100n GND
C17 100n GND
HUB_1V8PLL
C8 4.7u GND
C10 100n GND
HUB_1V8
C7 4.7u GND
C9 100n GND

U1
USB2517I-JZX
+3V3
VDD33 46
VDD33PLL 64
VDD33CR 24
VDDA33 10
VDDA33 52
VDDA33 57
VDDA33 35
HUB_1V8
HUB_1V8PLL
VDD18 25
VDD18PLL 62

UP_DP 59 USBUP_DP
UP_DN 58 USBUP_DM
XI 61 XTAL1/CLKIN
XO 60 XTAL2
RBIAS 63 RBIAS
HUB_RST_N 43 RESET_N
44 VBUS_DET
+3V3
GND 19 TEST
HUB_CFG0 41 SCL/SMBCLK/CFG_SEL0
HUB_CFG1 42 HS_IND/CFG_SEL1
HUB_CFG2 13 CFG_SEL2
HUB_NONREM1 40 SDA/SMBDATA/NON_REM1
HUB_NONREM0 45 SUSP_IND/LOCAL_PWR/NON_REMO
HUB_GANG 34 LED_B3_N/GANG_EN
HUB_BOOST0 50 LED_B1_N/BOOST0
HUB_BOOST1 48 LED_B2_N/BOOST1

USBDN1_DP/PRT_DIS_P1 2 PORT1_DP
USBDN1_DM/PRT_DIS_M1 1 PORT1_DN
USBDN2_DP/PRT_DIS_P2 4 PORT2_DP
USBDN2_DM/PRT_DIS_M2 3 PORT2_DN
USBDN3_DP/PRT_DIS_P3 7 PORT3_DP
USBDN3_DM/PRT_DIS_M3 6 PORT3_DN
USBDN4_DP/PRT_DIS_P4 9 PORT4_DP
USBDN4_DM/PRT_DIS_M4 8 PORT4_DN
USBDN5_DP/PRT_DIS_P5 12 CP_HUB_DP
USBDN5_DM/PRT_DIS_M5 11 CP_HUB_DN
USBDN6_DP/PRT_DIS_P6 54
USBDN6_DM/PRT_DIS_M6 53
USBDN7_DP/PRT_DIS_P7 56
USBDN7_DM/PRT_DIS_M7 55

PRTPWR1 29
OCS1_N 28
PRTPWR2 27
OCS2_N 26
PRTPWR3 23
OCS3_N 22
PRTPWR4 21
OCS4_N 20
PRTPWR5 17
OCS5_N 16
PRTPWR6 15
OCS6_N 14
PRTPWR7 31
OCS7_N 30

VSS 65
GND
LED_A1_N/PRT_SWP1 51
LED_A2_N/PRT_SWP2 49
LED_A3_N/PRT_SWP3 47
LED_A4_N/PRT_SWP4 33
LED_A5_N/PRT_SWP5 31
LED_A6_N/PRT_SWP6 11
LED_A7_N 18
LED_B4_N 39
LED_B5_N 38
LED_B6_N 37
LED_B7_N 36

R10 1M XO XI
Y1 X322524MSB4SI XI XO
C4 33p GND
C5 33p GND

U5
MAX809S
+3V3
VCC 3
RESET 2 HUB_RST_N
GND 1
GND
C6 100n GND

RBIAS R1 12k GND
HUB_SWP3 R3 10k GND
HUB_NONREM0 R4 10k GND
HUB_SWP4 R5 10k GND
HUB_SWP5 R9 10k GND
HUB_SWP1 HUB_SWP2 HUB_BOOST0 HUB_BOOST1 RN1 10k GND
HUB_SWP1 HUB_SWP2 HUB_SWP3 HUB_SWP4 HUB_SWP5 HUB_SWP6 HUB_SWP7 HUB_BOOST0 HUB_BOOST1 HUB_CFG0 HUB_CFG1 HUB_CFG2 HUB_NONREM0 HUB_NONREM1 HUB_GANG
HUB_GANG HUB_NONREM1 HUB_CFG0 HUB_CFG1 RN2 10k GND +3V3
HUB_SWP6 HUB_SWP7 HUB_CFG2 RN3 10k GND +3V3

Two TCA6408A expanders, one per pair of ports: EN, FAULT_N, RESET, BOOT

+3V3
C18 100n GND
U2
TCA6408ARGTR
+3V3
SCL 12
SDA 13
INT 11
EXP1_RST_N 1 RESET
ADDR 16 GND
Vcc 15
Vcca 14
EPAD 17
GND 6
GND
P0 2 PORT1_RST_IO
P1 3 PORT1_BOOT_IO
P2 4 PORT1_EN
P3 5 PORT1_FLT_N
P4 7 PORT2_EN
P5 8 PORT2_FLT_N
P6 9 PORT2_BOOT_IO
P7 10 PORT2_RST_IO
+3V3
R11 10k
EXP1_RST_N

+3V3
C19 100n GND
U6
TCA6408ARGTR
+3V3
SCL 12
SDA 13
INT 11
EXP2_RST_N 1 RESET
+3V3 16 ADDR
Vcc 15
Vcca 14
EPAD 17
GND 6
GND
P0 2 PORT3_RST_IO
P1 3 PORT3_BOOT_IO
P2 4 PORT3_EN
P3 5 PORT3_FLT_N
P4 7 PORT4_EN
P5 8 PORT4_FLT_N
P6 9 PORT4_BOOT_IO
P7 10 PORT4_RST_IO
+3V3
R12 10k
EXP2_RST_N

I²C isolation: open with VIN off, so the node's pull-ups cannot feed this board

+3V3
LEAD_SDA
Q1 2N7002
+3V3 1
SDA 2
R6 4.7k
SDA
LEAD_SCL
Q2 2N7002
+3V3 1
SCL 2
R7 4.7k
SCL

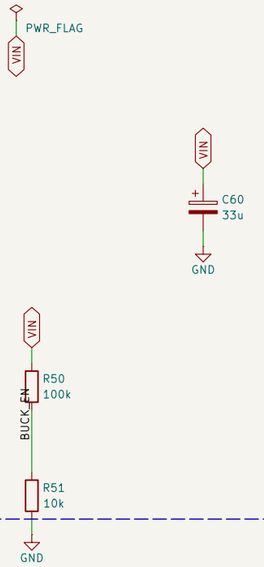
Hub reset: U5 (MAX809S, 2.93 V, ≥ 150 ms, push-pull) on RESET_N (errata DS80000628A, Module 2).
Hub: CFG_SEL[2:0] = 110 (internal default configuration, strap options disabled: DS00001598C Table 5-2):
self-powered, ports switched and sensed one by one (here by the expanders), no port disabled, D+ / D- not sampled at reset; the other strap pins keep their pull-downs, unread. Ports 6 and 7 enabled, unused (cosmetic). On purpose: PRTPWR unused and OCS_N on their pull-ups (the node switches each port and reads its FAULT_N through the expanders); VBUS_DET on 3.3 V (self-powered hub).
Hub port n = USB-C n (busid 1-1.n) for n = 1-4; port 5 = the co-processor.
Expander pins are inputs at power-up: the EN pull-downs keep every port off until the node writes.
No INT on the lead (J1.1 open): the node polls the expanders and the INA226s every 10-20 ms.
I²C: the node's pull-ups on LEAD_*. 4.7k / 4.7k here; with VIN off the chips here lose their registers: the node sets them up again after each VIN on, and leaves this board's addresses alone while off.
J1 (through-hole) stands on the rear face: soldered by hand (the PCBA places the front face only).

Generated from electronics/data-port-x4-v2/design.py — edit the Python, then regenerate
A converted bay's drawer board: four USB-C behind a hub, 32 switchable pins on 8 buses, a logic an
hw-testing-rack
Sheet: /Hub/
File: hub.kicad_sch

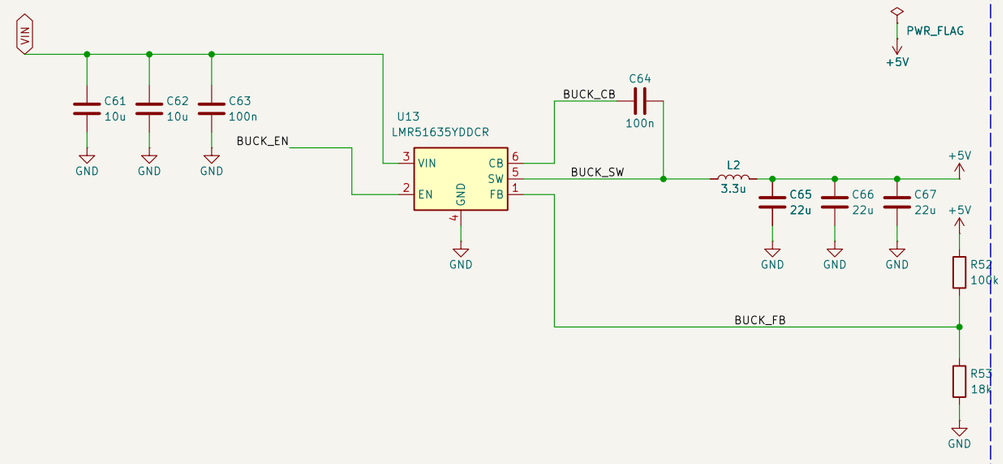
Title: DATA x4 v2 — Lead, hub, expanders, 3.3 V

Size: A2 | Date: | **Rev: 2.0**
KiCad E.D.A. 10.0.6 | Id: 2/9

VIN: the rack's 20 V, switched by the node (its eFuse), from the PH 8 lead's pin 8 (J1, sheet Hub)



+5V: LMR51635, synchronous, 1.1 MHz, 3.5 A (the DATA ×6's block, TI SLUSF64B table 8–2)



The DATA ×6's converter with the family's 3.5 A part (TI SLUSF64B table 8–2, LMR51635, 5 V, 1.1 MHz: 3.3 µH
Vishay IHLP2525CZ, 75 V, 13.5 A sat; 2 × 10 µF + 100 nF in, 3 × 22 µF out), PFM and spread spectrum as the
×6's LMR51625.
+5V: the four ports, the 3.3 V LDO (≤ 0.46 A) and the matrix: 2.46 A at 0.5 A a port; at worst, all four at
the TPS2553s' 0.644 A limit, 3.04 A, under the LMR51635's 3.5 A with no help from the node's firmware.
The inductor then peaks at 3.63 A from 28 V (1.17 A ripple), 3.93 A at the worst corner (L −20 %, 0.99 MHz),
under the LMR51635's 4.2 A minimum peak limit (SLUSF64B §6.5: 4.2 / 5.1 / 6.0 A; valley 3.2 / 4.0 / 4.8 A);
from 13.5 V: peak 3.48 A, valley 2.60 A. No frequency foldback (tON 170 ns ≥ 70, tOFF 556 ns ≥ 200).
FB 100k/18k: 5.24 V, as the node's. EN 100k/10k: on above 13.5 V, off below 10.8 V.
VIN on J1.8, the node's VRACK through its eFuse (TPS26600, off by default): no second protection here.
At the 3.04 A worst case (η ≈ 88 %): 0.91 A at 20 V, 1.21 A at 15 V, 1.34 A at 13.5 V, 0.65 A at 28 V;
the converter's full 3.5 A: 1.39 A at 15 V. Under the node eFuse's 1.76 A; a PH contact carries 2 A.

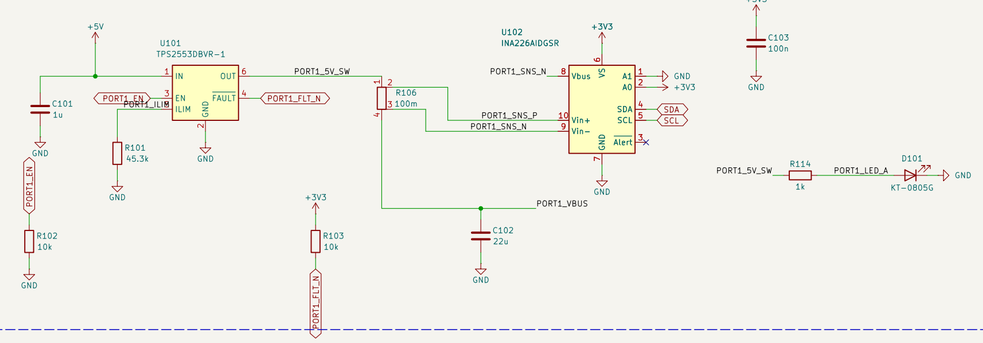
Generated from electronics/data-port-x4-v2/design.py — edit the Python, then regenerate
A converted bay's drawer board: four USB-C behind a hub, 32 switchable pins on 8 buses, a logic an
hw-testing-rack

Sheet: /Power/
File: power.kicad_sch

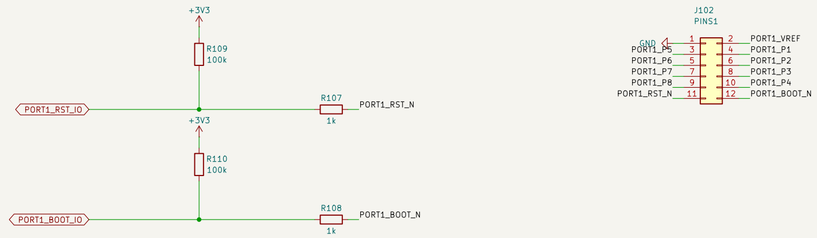
Title: DATA ×4 v2 — 20 V input, the 5 V converter

Size: A3	Date:	**Rev: 2.0**
KiCad E.D.A. 10.0.6		Id: 3/9

Port 1 power: +5V → TPS2553-1 → PORT1_5V_SW → 100 mΩ shunt → PORT1_VBUS, the INA226 across the shunt

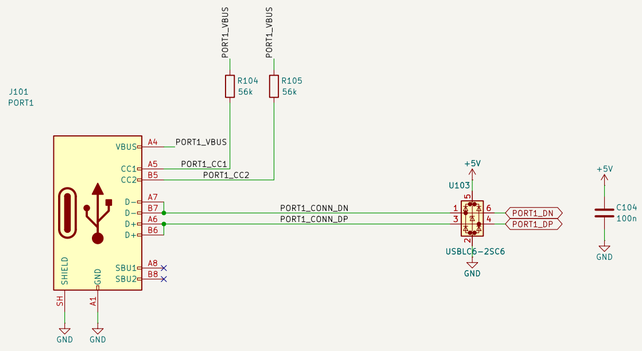


Header 1 (2 × 6 through-hole): GND, VREF, P1-P8, RESET_N, BOOT_N; RESET/BOOT through 1k, their expander lines pulled up

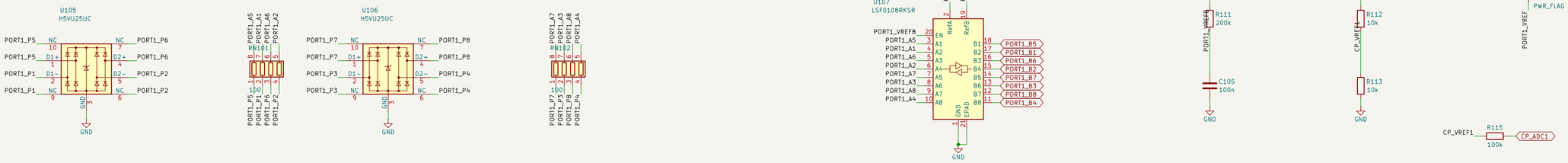


Port 1 = sub-slot A1.1, hub port 1, busid 1-1.1.
INA226 at 0x41: expander U2 (0x20).
ILIM 45.3k: 524 / 577 / 638 mA min / typ / max (TI SLVS841G §8.5.1 eq. 1-3), 519-644 mA with the
  resistor's 1 % tolerance; the −1 latches off after 7.5 ms in limit. FAULT_N flags it.
INA226 Vbus on the shunt's load-side sense pad (= VBUS).
EN pulled down: the port stays off until the node switches it on.
Pins: P1-P8 → crosspoint X lines (docs/electronics.md › DATA ×4 v2 map);
  an unwired VREF stays ≈ 0.4 V, under the LSF0108's 0.65 V: the firmware keeps the port's
  crosspoint switches open while its VREF reads under 0.65 V (GPIO40-43 under 0.33 V).
ESD: VREF through U108, RESET_N / BOOT_N through U109 (flow-through).
VREF: the board's I/O supply, 0.95-3.6 V; a 5 V board with TTL inputs: its own 3.3 V pin.
Source only (Rp from the switched VBUS); a back-feeding source trips the TPS2553 (latched).
The DUT's RESET/BOOT pull-ups ≥ 4.7k: against 1k, the 1k in series stops at ≈ 1.65 V.
VBUS LED: on while the port's switch is on (5V_SW, before the shunt).

USB-C 1 (source): Rp 56k on each CC; USBLC6 in flow-through (connector side | hub side)



Pins 1: ESD in flow-through, 100 Ω, LSF0108 (A = the DUT's VREF, B = 3.3 V) to crosspoint U7's X lines



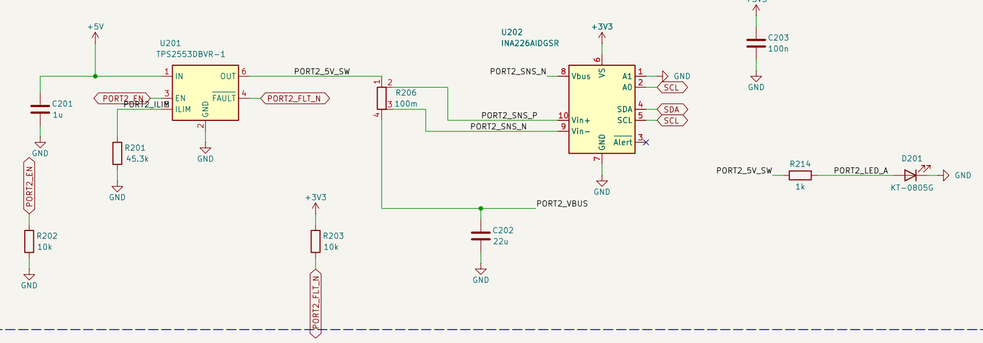
ESD on the header's VREF (U108, in flow-through: header → ESD → LSF0108 and divider) and on RESET_N / BOOT_N (U109: header → ESD → 1k)

Generated from electronics/data-port-x4-v2/design.py — edit the Python, then regenerate
A converted bay's drawer board: four USB-C behind a hub, 32 switchable pins on 8 buses, a logic an
hw-testing-rack
Sheet: /Port 1/
File: port1.kicad_sch

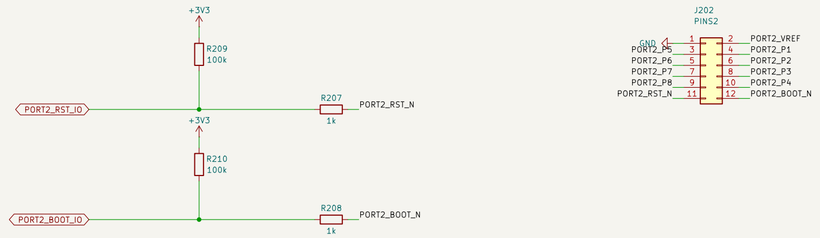
Title: DATA ×4 v2 — USB-C 1: switch, current sense, ESD, header; pins

Size: A2	Date:	**Rev: 2.0**
KiCad E.D.A. 10.0.6		Id: 4/9

Port 2 power: +5V → TPS2553-1 → PORT2_5V_SW → 100 mΩ shunt → PORT2_VBUS, the INA226 across the shunt

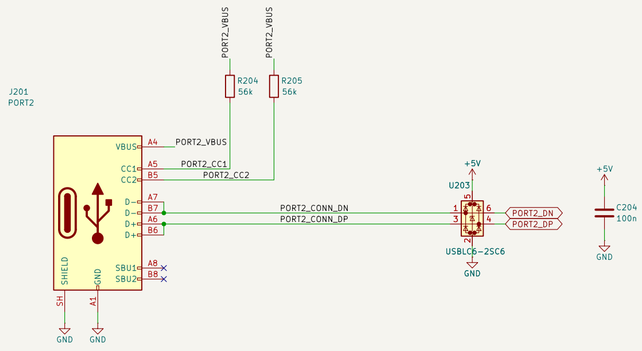


Header 2 (2 × 6 through-hole): GND, VREF, P1-P8, RESET_N, BOOT_N; RESET/BOOT through 1k, their expander lines pulled up



Port 2 = sub-slot A1.2, hub port 2, busid 1-1.2.
INA226 at 0x43; expander U2 (0x20).
ILIM 45.3k: 524 / 577 / 638 mA min / typ / max (TI SLVS841G §8.5.1 eq. 1-3), 519-644 mA with the resistor's 1 % tolerance; the ~1 latches off after 7.5 ms in limit. FAULT_N flags it.
INA226 Vbus on the shunt's load-side sense pad (= VBUS).
EN pulled down: the port stays off until the node switches it on.
Pins: P1-P8 → crosspoint X lines (docs/electronics.md › DATA ×4 v2 map); an unwired VREF stays ≈ 0.4 V, under the LSF0108's 0.65 V: the firmware keeps the port's crosspoint switches open while its VREF reads under 0.65 V (GPIO40-43 under 0.33 V).
ESD: VREF through U208, RESET_N / BOOT_N through U209 (flow-through).
VREF: the board's I/O supply, 0.95-3.6 V; a 5 V board with TTL inputs: its own 3.3 V pin. Source only (Rp from the switched VBUS); a back-feeding source trips the TPS2553 (latched).
The DUT's RESET/BOOT pull-ups ≥ 4.7k: against 1k, the 1k in series stops at ≈ 1.65 V.
VBUS LED: on while the port's switch is on (5V_SW, before the shunt).

USB-C 2 (source): Rp 56k on each CC; USBLC6 in flow-through (connector side | hub side)



Pins 2: ESD in flow-through, 100 Ω, LSF0108 (A = the DUT's VREF, B = 3.3 V) to crosspoint U7's X lines



ESD on the header's VREF (U208, in flow-through: header → ESD → LSF0108 and divider) and on RESET_N / BOOT_N (U209: header → ESD → 1k)

Generated from electronics/data-port-x4-v2/design.py — edit the Python, then regenerate.
A converted bay's drawer board: four USB-C behind a hub, 32 switchable pins on 8 buses, a logic an

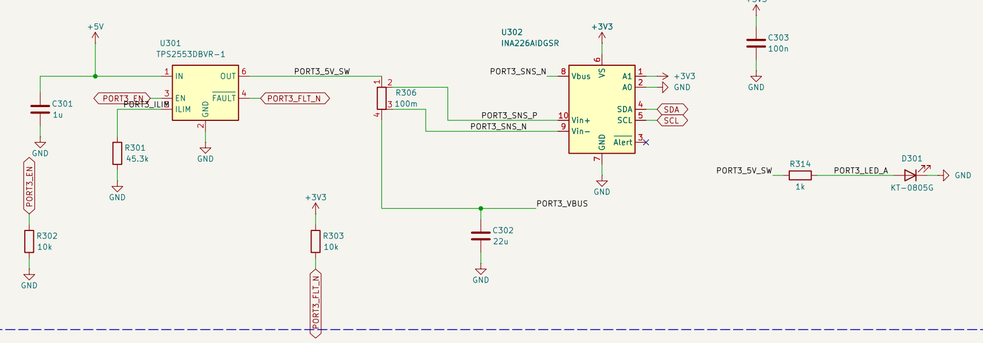
hw-testing-rack

Sheet: /Port 2/
File: port2.kicad_sch

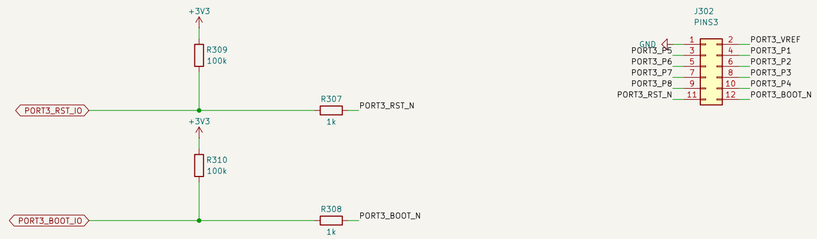
Title: DATA ×4 v2 — USB-C 2: switch, current sense, ESD, header; pins

Size: A2	Date:	**Rev: 2.0**
KiCad E.D.A. 10.0.6		Id: 5/9

Port 3 power: +5V → TPS2553-1 → PORT3_5V_SW → 100 mΩ shunt → PORT3_VBUS, the INA226 across the shunt

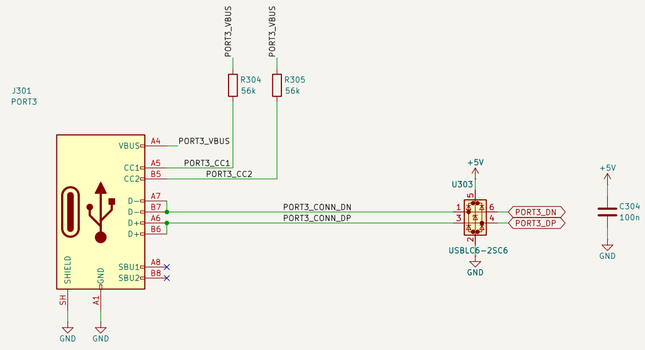


Header 3 (2 × 6 through-hole): GND, VREF, P1-P8, RESET_N, BOOT_N; RESET/BOOT through 1k, their expander lines pulled up

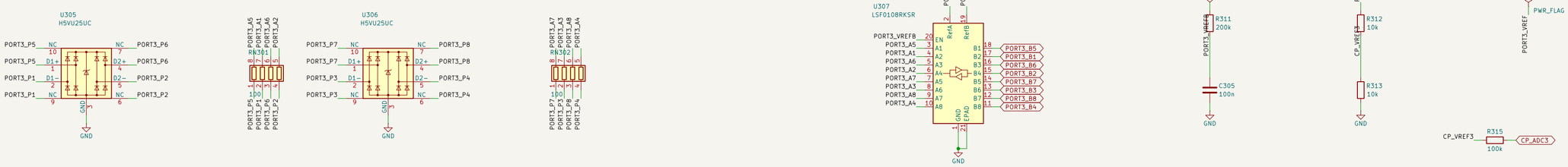


Port 3 = sub-slot A1.3, hub port 3, busid 1-1.3.
INA226 at 0x44; expander U6 (0x21).
ILIM 45.3k: 524 / 577 / 638 mA min / typ / max (TI SLVS841G §8.5.1 eq. 1-3), 519-644 mA with the
  resistor's 1 % tolerance; the −1 latches off after 7.5 ms in limit. FAULT_N flags it.
INA226 Vbus on the shunt's load-side sense pad (= VBUS).
EN pulled down: the port stays off until the node switches it on.
Pins: P1-P8 → crosspoint X lines (docs/electronics.md › DATA ×4 v2 map);
  an unwired VREF stays ≈ 0.4 V, under the LSF0108's 0.65 V: the firmware keeps the port's
  crosspoint switches open while its VREF reads under 0.65 V (GPIO40-43 under 0.33 V).
ESD: VREF through U308, RESET_N / BOOT_N through U309 (flow-through).
VREF: the board's I/O supply, 0.95-3.6 V; a 5 V board with TTL inputs: its own 3.3 V pin.
Source only (Rp from the switched VBUS); a back-feeding source trips the TPS2553 (latched).
The DUT's RESET/BOOT pull-ups ≥ 4.7k: against 1k, the 1k in series stops at ≈ 1.65 V.
VBUS LED: on while the port's switch is on (5V_SW, before the shunt).

USB-C 3 (source): Rp 56k on each CC; USBLC6 in flow-through (connector side | hub side)



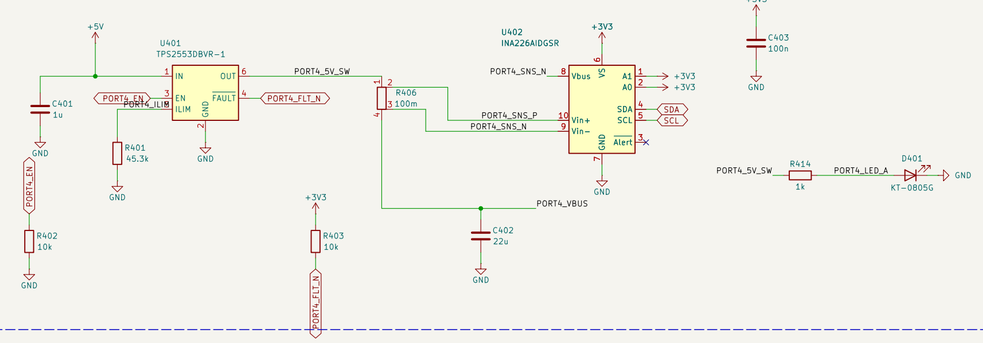
Pins 3: ESD in flow-through, 100 Ω, LSF0108 (A = the DUT's VREF, B = 3.3 V) to crosspoint U8's X lines



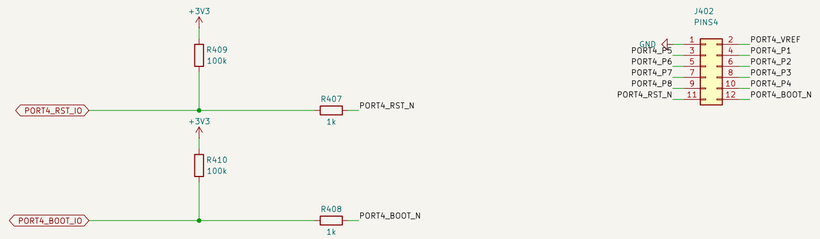
ESD on the header's VREF (U308, in flow-through: header → ESD → LSF0108 and divider) and on RESET_N / BOOT_N (U309: header → ESD → 1k)

Generated from electronics/data-port-x4-v2/design.py — edit the Python, then regenerate
A converted bay's drawer board: four USB-C behind a hub, 32 switchable pins on 8 buses, a logic an
hw-testing-rack
Sheet: /Port 3/
File: port3.kicad_sch
Title: DATA ×4 v2 — USB-C 3: switch, current sense, ESD, header; pins
Size: A2 | Date: | **Rev: 2.0**
KiCad E.D.A. 10.0.6 | Id: 6/9

Port 4 power: +5V → TPS2553-1 → PORT4_5V_SW → 100 mΩ shunt → PORT4_VBUS, the INA226 across the shunt

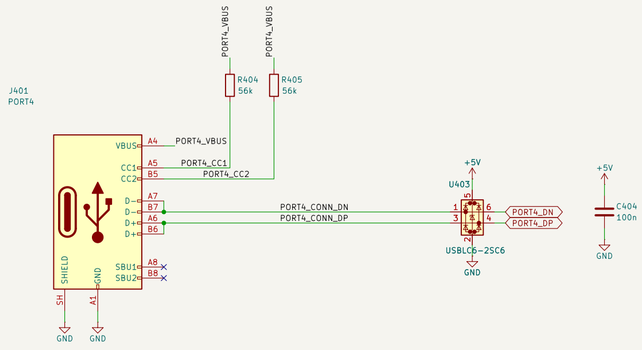


Header 4 (2 × 6 through-hole): GND, VREF, P1-P8, RESET_N, BOOT_N; RESET/BOOT through 1k, their expander lines pulled up

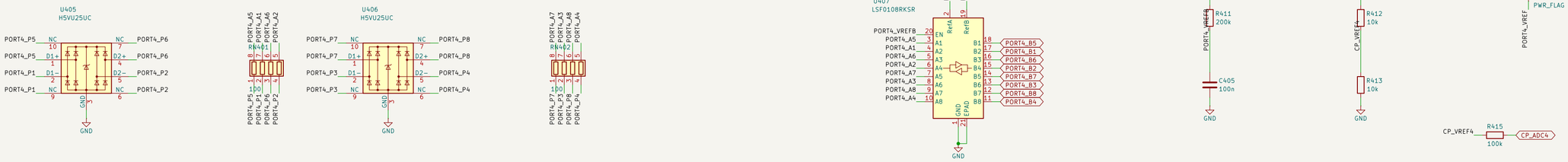


Port 4 = sub-slot A1.4, hub port 4, busid 1-1.4.
INA226 at 0x45; expander U6 (0x21).
ILIM 45.3k: 524 / 577 / 638 mA min / typ / max (TI SLVS841G §8.5.1 eq. 1-3), 519-644 mA with the resistor's 1 % tolerance; the −1 latches off after 7.5 ms in limit. FAULT_N flags it.
INA226 Vbus on the shunt's load-side sense pad (= VBUS).
EN pulled down: the port stays off until the node switches it on.
Pins: P1-P8 → crosspoint X lines (docs/electronics.md › DATA ×4 v2 map); an unwired VREF stays ≈ 0.4 V, under the LSF0108's 0.65 V: the firmware keeps the port's crosspoint switches open while its VREF reads under 0.65 V (GPIO40-43 under 0.33 V).
ESD: VREF through U408, RESET_N / BOOT_N through U409 (flow-through).
VREF: the board's I/O supply, 0.95-3.6 V; a 5 V board with TTL inputs: its own 3.3 V pin. Source only (Rp from the switched VBUS); a back-feeding source trips the TPS2553 (latched).
The DUT's RESET/BOOT pull-ups ≥ 4.7k: against 1k, the 1k in series stops at ≈ 1.65 V.
VBUS LED: on while the port's switch is on (5V_SW, before the shunt).

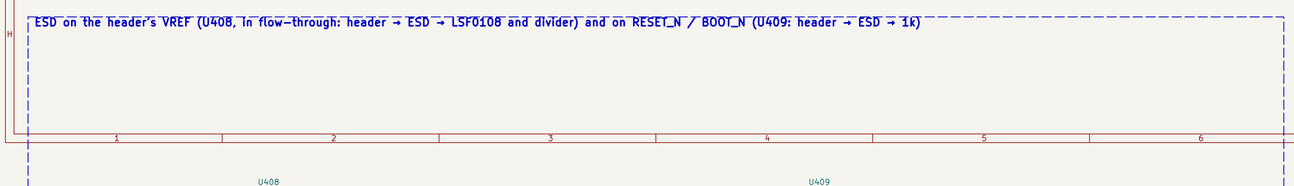
USB-C 4 (source): Rp 56k on each CC; USBLC6 in flow-through (connector side | hub side)



Pins 4: ESD in flow-through, 100 Ω, LSF0108 (A = the DUT's VREF, B = 3.3 V) to crosspoint U8's X lines



ESD on the header's VREF (U408, in flow-through: header → ESD → LSF0108 and divider) and on RESET_N / BOOT_N (U409: header → ESD → 1k)



Generated from electronics/data-port-x4-v2/design.py — edit the Python, then regenerate.
A converted bay's drawer board: four USB-C behind a hub, 32 switchable pins on 8 buses, a logic an

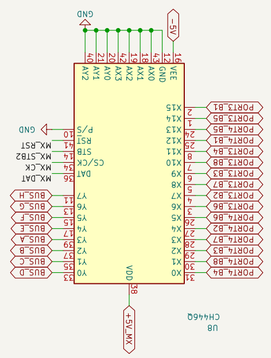
hw-testing-rack

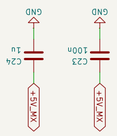
Sheet: /Port 4/
File: port4.kicad_sch

Title: DATA ×4 v2 — USB-C 4: switch, current sense, ESD, header; pins

Size: A2	Date:	**Rev: 2.0**
KiCad E.D.A. 10.0.6		Id: 7/9

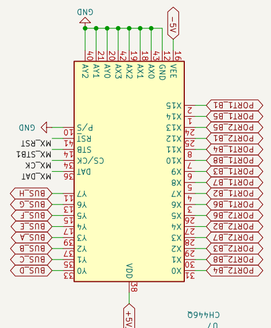
Crosspoint U8 (CH446Q at ±5 V, serial addressing): X = ports 3-4's pins, Y0-Y7 = buses DCBAEFGH

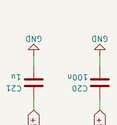






Crosspoint U7 (CH446Q at ±5 V, serial addressing): X = ports 1-2's pins, Y0-Y7 = buses DCBAEFGH







The node's matrix expander (TCA9534A, 0x38): the crosspoints and the co-processor's RUN / BOOTSEL

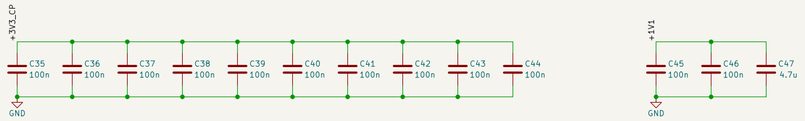
CH446Q serial mode (P/S low): 7 address bits on CK, data on DAT, latched by STB; RST high = all open.
TCA9534A 0x38: P0 DAT, P1 CK, P2 STB, P3 RST, P4 RUN, P5 BOOTSEL, P7 mark.
RUN and BOOTSEL are open drain emulated (output register 0, the pin switched to output).
±5 V on purpose: at a single 5 V R_on is 120-225 Ω, and a 1.8 V board's I²C low would fail.

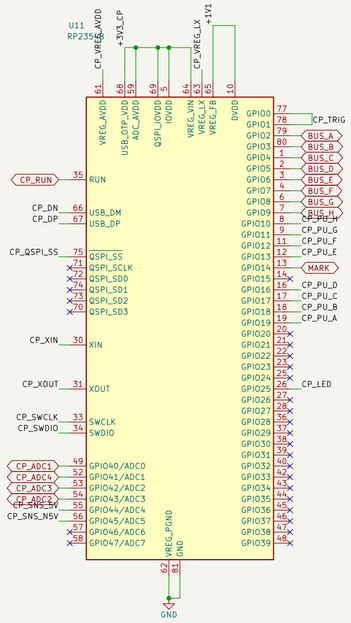
+5V_MX through a ferrite (the crosspoints, the inverter, the LSF0108' bias); -5 V for the crosspoints' VEE (TPS60400 inverter)

Generated from electronics/data-port-x4-v2/design.py — edit the Python, then regenerate.
A converted bay's drawer board: four USB-C behind a hub, 32 switchable pins on 8 buses, a logic an

Sheet: /Matrix/	
File: matrix.kicad_sch	
Title: DATA ×4 v2 — Crosspoints, −5 V, the node's matrix expander	
Size: A2	Date:
KiCad E.D.A. 10.0.6	Id: 8/9
hw-testing-rack	**Rev: 2.0**

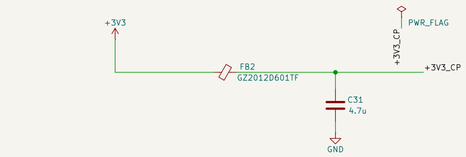
RP2354B, wired like a Pico 2 (gusmanb's PICO_2 UF2 runs as it is); its decoupling

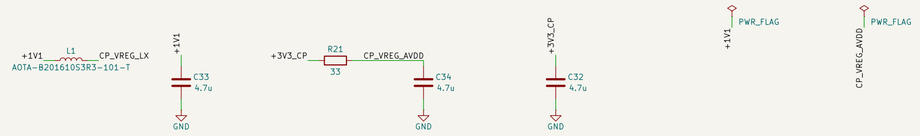




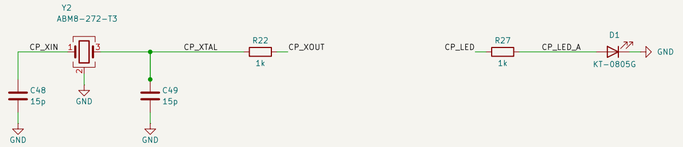
GPIO2-9 = buses A-H (gusmanb CH1-8, 8-bit mode); GPIO0 → GPIO1 bridged (its trigger path).
GPIO14 = the node's mark (TCA9534A P7); GPIO25 = LED; GPIO19…16, 13…10 = pull-up enables A…H (2.2k);
GPIO40, 43, 42, 41 = VREF 1:4 / 2 (through 100k); GPIO44 = +5V × 47/147;
GPIO45 = 3.3 V − (3.3 + 5) × 10/110 from −5 V (2.55 V nominal).
RUN pulled up, BOOTSEL (QSPI_SS) through 1k: both held by the node (TCA9534A), open drain.
Core regulator: Raspberry Pi's Minimal R4 block turned 180°: L1's dot (pad 1) on +1V1, not VREG_LX.
C47 (DVDD pin 32, furthest from the regulator) 4.7 µF: no copper under LX / L1 (In1, In2, B.Cu).

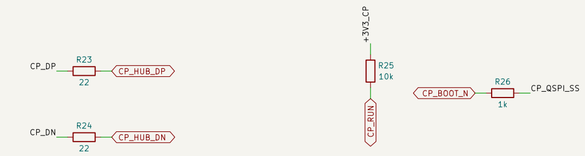
Its 3.3 V from +3V3 through a ferrite, the core regulator as Raspberry Pi lays it out (3.3 µH, 3 × 4.7 µF, 33 Ω)



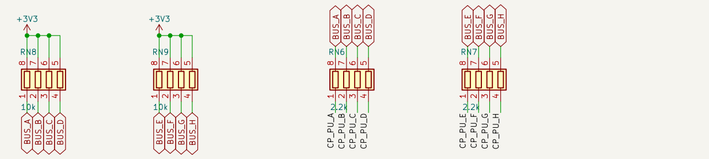


12 MHz crystal (ABM8-272-T3, 15 pF, 1k on XOUT), USB on hub port 5 (22 Ω), RUN, BOOTSEL, LED, SWD





Bus pull-ups: 10k to 3.3 V always (the LSFs' B side), 2.2k switched by the co-processor (GPIO10-13, 16-19); the 15 V rails into the ADC



Generated from electronics/data-port-x4-v2/design.py — edit the Python, then regenerate.
A converted bay's drawer board: four USB-C behind a hub, 32 switchable pins on 8 buses, a logic an

hw-testing-rack		
Sheet: /Co-processor/		
File: coproc.kicad_sch		
Title: DATA ×4 v2 — RP2354B analyzer / co-processor on hub port 5		
Size: A2	Date:	**Rev: 2.0**
KiCad E.D.A. 10.0.6		Id: 9/9